Computers in Biology and Medicine 195 (2025) 110647

Contents lists available at ScienceDirect

Computers in Biology and Medicine

journal homepage: www.elsevier.com/locate/compbiomed

Classification of knee osteoarthritis severity using markerless motion capture and long short-term memory fully convolutional network

Engin Kaya [a], Hülya Şirzai [b], Güneş Yavuzer [b], Hande Argunsah [c,*]

[a] Acibadem Mehmet Ali Aydinlar University, Institute of Natural Sciences, Department of Biomedical Engineering, Istanbul, Turkey
[b] Romatem Physical Therapy and Rehabilitation Hospital, Istanbul, Turkey
[c] Acibadem Mehmet Ali Aydinlar University, Faculty of Engineering and Natural Sciences, Department of Biomedical Engineering, Istanbul, Turkey

ARTICLE INFO

Keywords:
Knee osteoarthritis
Gait analysis
Deep learning
LSTM-FCN
Markerless motion capture
Classification
Feature extraction
Gait cycle detection

ABSTRACT

This study explored the integration of markerless motion capture and deep learning to classify knee osteoarthritis severity based on gait kinematics, providing an alternative to traditional assessment methods. We employed a Long Short-Term Memory Fully Convolutional Network model to analyze gait patterns and distinguish severity levels corresponding to Kellgren–Lawrence grades. Two data-splitting strategies were compared: random splitting, which allowed learning from a diverse set of individuals, and subject-based splitting, which assessed the model's ability to generalize to unseen patients. The model achieved high classification performance in the random split (accuracy = 0.91), but performance declined in the subject-based split (accuracy = 0.76), indicating challenges in generalizing across individuals. Severe and healthy groups were well classified, while early and moderate severity groups showed higher misclassification rates due to overlapping gait characteristics. Our findings highlight the potential of gait-based deep learning models for automated severity classification, offering a scalable and accessible alternative to conventional assessments. However, challenges related to inter-subject variability suggest the need for enhanced feature extraction, multimodal data integration, and domain adaptation to improve generalizability. Future research should focus on longitudinal data to assess the model's predictive capability for disease progression and treatment outcomes.

1. Introduction

Knee osteoarthritis (OA) is a leading cause of pain, disability, and reduced quality of life among older adults worldwide. Characterized by progressive degeneration of articular cartilage, subchondral bone remodeling, and synovial inflammation, knee OA often results in impaired joint function and altered gait mechanics [1,2]. The severity of knee OA is assessed using the Kellgren–Lawrence grading system, which classifies the condition based on radiographic features such as joint space narrowing, osteophyte formation, and subchondral sclerosis [3]. While radiographic evaluations provide valuable structural information, they fail to capture functional deficits and dynamic movement patterns that may distinguish individuals with knee OA from healthy individuals [4]. Moreover, they primarily focus on static structural changes, such as joint space narrowing and osteophyte formation, without capturing functional impairments or dynamic movement patterns essential for evaluating mobility. Radiographic changes often appear only in later stages, missing early cartilage and soft tissue abnormalities that could enable early intervention. Structural findings on radiographs may also poorly correlate with symptoms, complicating treatment planning. Furthermore, radiographs provide only static snapshots, failing to monitor dynamic disease progression, and repeated imaging poses cumulative radiation risks. Magnetic resonance imaging (MRI) is increasingly employed to evaluate knee OA, providing detailed insights into soft tissue abnormalities, cartilage degradation, and bone marrow lesions [5]. Moreover, biochemical markers measured in synovial fluid, blood, and urine have been explored as potential indicators of disease progression, complementing imaging techniques by identifying molecular changes associated with knee OA [6]. Despite the availability of these methods, integrating structural and functional assessments through non-invasive, affordable, and rapid methods remains a challenge in routine practice.

Quantitative gait analysis has emerged as an alternative tool for evaluating biomechanical alterations associated with knee OA non-invasively. Previous studies using marker-based motion capture systems have reported kinematic abnormalities, including reduced knee

* Corresponding author. Acibadem Mehmet Ali Aydinlar University, Icerenkoy Mah. Kayisdagi Cad. No:32 Atasehir, Istanbul Turkey.
E-mail address: hande.argunsah@gmail.com (H. Argunsah).

https://doi.org/10.1016/j.compbiomed.2025.110647
Received 10 March 2025; Received in revised form 10 June 2025; Accepted 17 June 2025
Available online 28 June 2025
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flexion, altered joint moments, and asymmetrical loading patterns during walking in individuals with knee OA [7,8]. However, these systems are often costly, time-intensive, and require extensive laboratory setups, limiting their clinical utility and accessibility [9].

Recent advancements in markerless motion capture technology offer a promising alternative for assessing gait biomechanics. These systems enable non-invasive, efficient, and cost-effective evaluation of kinematic parameters, making them well-suited for large-scale studies and clinical applications [10,11]. Markerless systems rely on computer vision algorithms and artificial intelligence to accurately track movement, eliminating the need for reflective markers and specialized attire. Despite the increasing application of markerless motion capture and machine learning in clinical biomechanics, there remains a significant gap in leveraging these tools for the automated classification of knee osteoarthritis severity. Previous studies have largely focused on either radiographic grading or kinematic assessment using marker-based systems, which, while effective, are resource-intensive and lack scalability for widespread clinical use [11,12]. Furthermore, few investigations have explored the application of deep learning models to raw gait kinematics obtained through markerless systems.

The primary objective of this study is to evaluate whether gait kinematics captured through a markerless motion capture pipeline can be effectively used to classify knee OA severity according to Kellgren–Lawrence grades. We aim to develop a robust deep learning model for automatic severity classification and assess the performance of the model under both random and subject-based data-splitting strategies. By addressing these aims, we seek to contribute a scalable and non-invasive method for functional OA assessment that bridges the gap between AI-driven biomechanics and clinical utility.

2. Materials and methods

In this study, knee osteoarthritis (OA) severity was classified based on gait kinematics using a Long Short-Term Memory Fully Convolutional Network (LSTM-FCN). Gait data were captured through a camera, which was used to extract 3D joint locations of the lower limbs using a markerless pose estimation algorithm. The gait cycles were detected from the joint locations and used for OA severity classification. Further details on the data processing, model architecture, and evaluation procedures are provided in the following sections.

2.1. Dataset

This study utilized gait data sourced from the publicly available dataset referenced in Ref. [13], entitled "A vision-based gait dataset for knee osteoarthritis and Parkinson's disease analysis with severity levels". Although the original dataset includes participants with either knee OA or Parkinson's disease (PD), only gait recordings from individuals with knee OA and healthy controls were used in this study. All subjects with PD were excluded. The gait data was collected in a hospital setting using a Nikon DSLR D5300 camera positioned 8 m from a 4.5 m walking mat with a green background. Six red-colored passive reflective markers were placed on key anatomical landmarks of the subjects before performing the gait. The dataset included gait sequences in the sagittal plane, with video recordings taken from both left-to-right and right-to-left perspectives. Each gait video was captured in MOV format at a frame rate of 50 fps and a resolution of 1920x1080, with recordings lasting approximately 10–13 s, yielding 400–650 frames per sequence.

Single-level walking recordings of participants diagnosed with knee osteoarthritis (KOA) and healthy control subjects (NM) were collected. Initially, knee X-rays of patients were conducted, and 50 subjects (38 females, 12 males) were diagnosed with KOA by orthopedic experts. Based on their X-ray reports, the patients were graded using the Kellgren–Lawrence scale (0–4 grades), with those in Grades 1–2 (early, EL), 3 (moderate, MD), and 4 (severe, SV) being included in the study. Patients with hip or other forms of osteoarthritis were excluded. The dataset comprised 50 KOA patients, categorized into early (n = 15), moderate (moderate = 20), and severe (severe = 15) stages of the disease. Additionally, 30 healthy individuals (17 females, 13 males) with no prior history of KOA or other orthopedic conditions were recruited as controls to enable comparisons between normal and pathological gait patterns.

2.2. Data processing

The dataset utilized in this study consists of publicly available video data. Rather than directly employing the raw video or the results of pose estimation, it was necessary to process the video to derive more informative and relevant data. To this end, the following steps illustrated in Fig. 1 were implemented as part of the data preprocessing and analysis pipeline.

2.2.1. Pose estimation

The pipeline of data processing steps used to predict the knee OA severity was represented in Fig. 1. Markerless motion capture techniques were employed to extract 3D joint locations from the gait recordings in the dataset. The videos included 4 distinct classes representing varying severity levels of knee OA and NM. To achieve high precision in pose estimation, a Hybrid Analytical-Neural Inverse Kinematics Solution for 3D Human Pose and Shape Estimation (HybrIK) [14] was utilized in combination with AlphaPose [15], a well-established framework for human pose estimation.

HybrIK combines model-based and key point-based approaches to achieve accurate 3D pose estimation and realistic body mesh reconstruction. By transforming 3D key points into body-part rotations using a novel twist-and-swing decomposition, it addresses limitations like image-model misalignment and unrealistic structures. This hybrid approach ensures pixel-aligned accuracy, outperforming state-of-the-art methods in 3D human pose and shape estimation benchmarks. We chose this method primarily because of its high accuracy [14] and the ease of implementation offered by the AlphaPose framework. The algorithm was applied to each frame of the gait videos to extract sequences of 3D joint locations, capturing 24 key points for each subject. Hence, we obtained data with the shape of [sequence length x 24 (number of key points) x 3 (coordinate dimensions)] for each gait recording. Notably, the HybrIK 3D pose estimation algorithm standardizes its outputs relative to the pelvis keypoint (x = 0, y = 0, z = 0) by default, ensuring a consistent reference frame across video sequences.

2.2.2. Gait cycle extraction

To extract gait cycles from the sequences of 3D joint locations, a preprocessing step was implemented to minimize noise in the data. Specifically, a one-dimensional uniform filter with a window size of 5 was applied to each dimension of the joint trajectories along the temporal axis. The filter window size of 5 was selected empirically based on visual assessments of trajectory smoothness and preservation of gait dynamics, representing a balance between noise reduction and signal integrity. This procedure facilitated the generation of smoother and more reliable joint trajectories, thereby enhancing the robustness of subsequent analyses. Following the smoothing process, a coordinate-based algorithm was adapted to identify key gait events using joint position data. Given that the dataset included ankle and pelvis joint locations but lacked specific foot markers, the algorithm was modified to detect heel strike (HS) events by evaluating the relative displacement between the ankle and pelvis joints [16].

HS events were identified as the peaks of the maximum differentiation between the ankle and pelvis joint positions along the sagittal plane. A gait cycle was subsequently defined as the interval between two consecutive HS events. This procedure was conducted independently for the right and left legs, ensuring precise segmentation of individual gait cycles. By integrating the smoothing step with this adapted approach, gait cycles were effectively delineated, accommodating subjects with

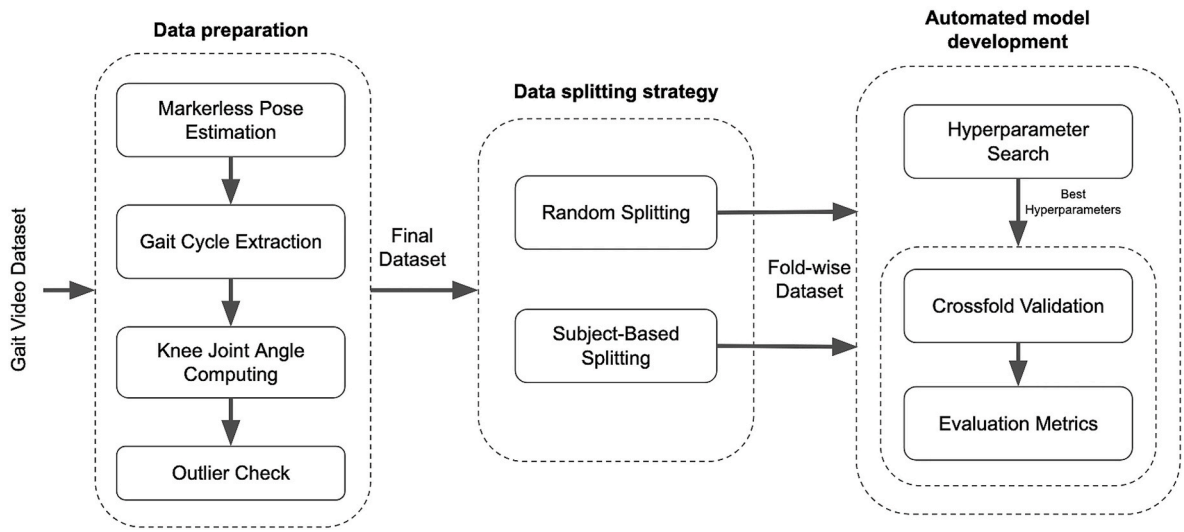

Fig. 1. Pipeline of data processing steps used to predict the knee OA severity.

diverse gait patterns as shown in Fig. 2.

2.2.3. Data cleaning

Given that the dataset does not specify which leg was affected by osteoarthritis (OA). Knee OA severity was graded based on radiographic evaluation of the most affected knee; however, since laterality was not specified in the dataset, gait cycles from both limbs were included in the analysis to ensure a comprehensive assessment of movement patterns and potential compensatory strategies. This approach allows us to analyze the overall gait dynamics without being limited by the absence of specific limb information. By utilizing data from both the affected and unaffected legs, we aim to capture a more holistic view of the gait cycle, reflecting potential compensatory mechanisms that may arise in response to the condition.

Although smoothing filters were applied to the joint locations, the incorporated gait cycle dataset may still exhibit noise resulting from variability in joint trajectories or inaccuracies in the pose estimation process. To enhance the quality of the dataset, a comprehensive multi-step noise elimination procedure was implemented. The initial step in the data refinement process involved filtering out noisy gait cycles based on their sequence lengths. Specifically, gait cycles containing fewer than 20 data points were removed, as these were likely incomplete or erroneous segments. For the remaining cycles, knee joint angles were computed using the positions of the hip, knee, and ankle joints. To ensure a uniform temporal representation across all samples, the calculated angles were resampled to 101 data points per gait cycle Fig. 3.

Following this preprocessing step, a statistical outlier detection algorithm was applied to the resampled data. This method utilized the mean and standard deviation of the knee joint angle trajectories within each target class. A tolerance threshold of two standard deviations was

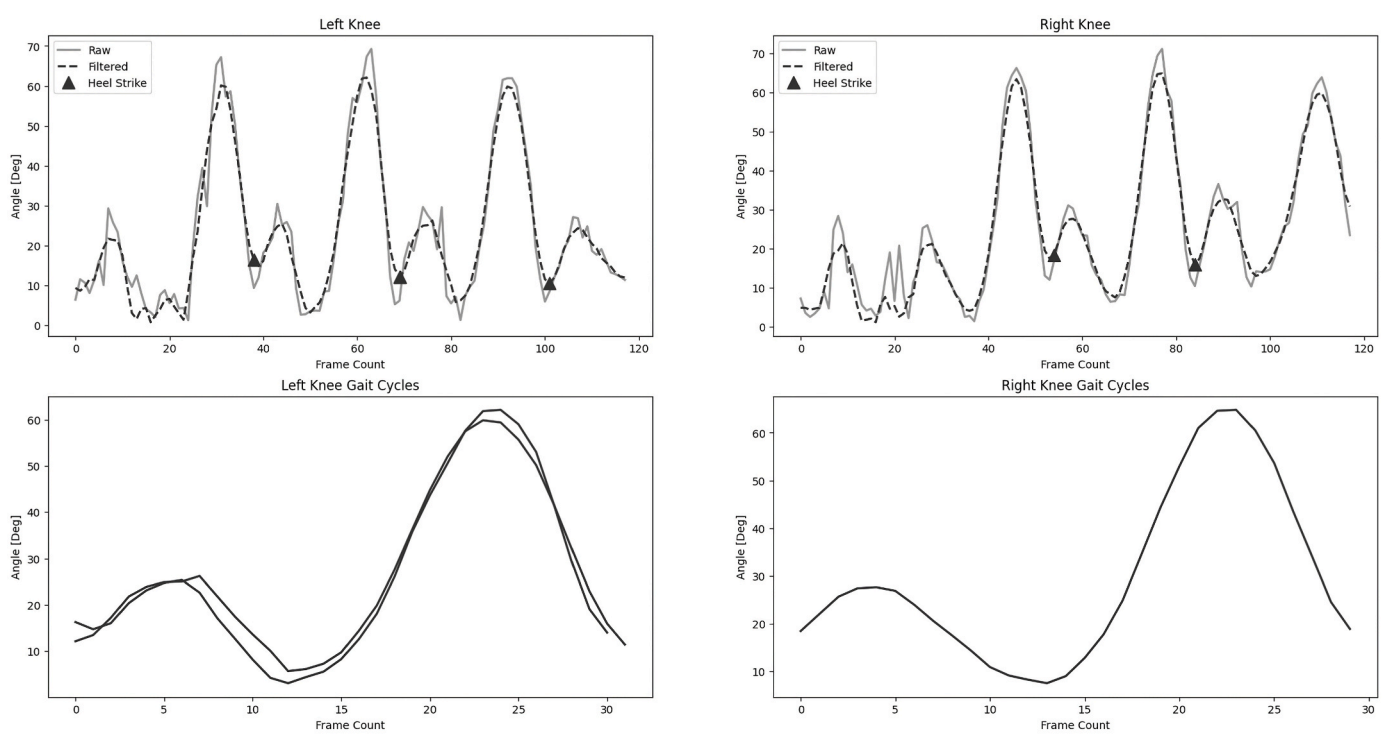

Fig. 2. Example of gait cycle extractions from a healthy control subject (NM-Subject 001-Video 01).

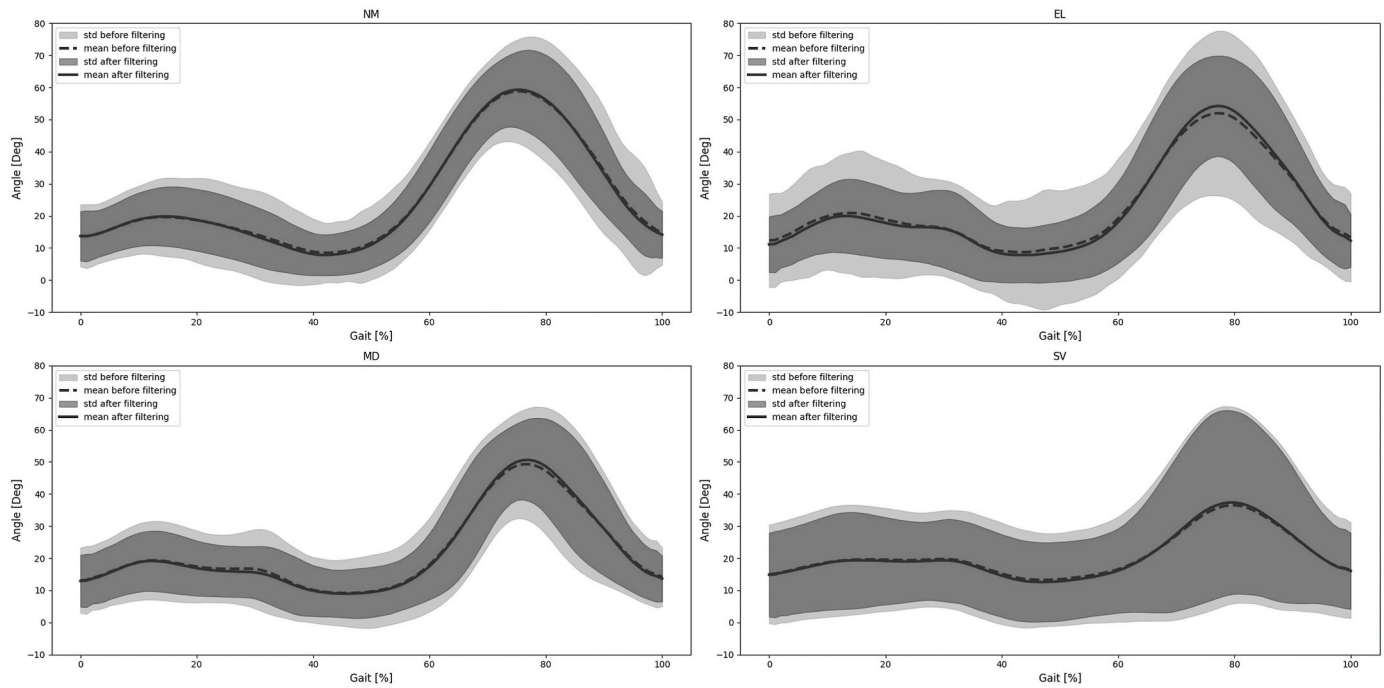

Fig. 3. Knee joint angle comparison of gait cycle data before and after filtering. The dashed lines represented the gait cycle before filtering, while the solid line represented the gait cycle after filtering. The black dashed and white solid lines represented the mean values, and the gray-shaded areas indicated the corresponding 1σ-uncertainties of gait cycles before and after filtering, respectively.

established to define acceptable data bounds. Gait cycles that exceeded these bounds in any dimension of the knee joint angle trajectory were identified as outliers and excluded from further analysis. This procedure was conducted separately for each target group and for both legs, accounting for gait cycles in both right-to-left and left-to-right walking directions. Knee joint angles were calculated solely for data cleaning purposes using geometric computations from the 3D positions of the hip, knee, and ankle joints in the sagittal plane. The data cleaning process we have implemented ensures the retention of high-quality gait cycle data, which is essential for analyzing gait patterns across different severity levels of knee OA and healthy controls.

To specifically address the analysis of knee OA, the final dataset was curated by retaining only the lower body joint locations. The selected joints included the left and right hip (l-hip, r-hip), knee (l-kne, r-kne), ankle (l-ank, r-ank), and foot (l-foot, r-foot) key points which are obtained from HybrIK predictions. Thus, only the 3D joint trajectories of these keypoints were used as input to the model, without incorporating any additional features such as clinical metadata or temporal annotations. These joint trajectories, initially normalized to a range between 0 and 1 by the 3D pose estimation algorithm, were subsequently resampled to 101 data points per gait cycle, ensuring a consistent temporal resolution throughout the dataset. Finally, we obtained data with the shape of [725 (sample) x 24 (feature size = number of key points x three dimensions) X 101 (sequence length)] for each gait cycle. The resulting dataset provided a standardized framework for subsequent analyses, facilitating the effective classification of gait patterns and enabling the differentiation of knee OA severity levels Table 1. As the KL grade was assigned based on the most affected knee without laterality information, the classification task in this study is defined as predicting the severity level of the most affected knee based on bilateral gait patterns. A total of 80 subjects contributed 1033 gait cycles, of which 725 remained after data cleaning. The SV group yielded the most useable gait data, as participants walked within a constrained area and exhibited shorter step lengths, whereas the EL group provided the least useable data. This cleaned dataset formed the basis for training and evaluating the classification model.

Table 1
Distribution of subjects, videos, and gait cycles across OA severity groups after preprocessing and data cleaning.

Target	Number of Subjects	Total Number of Videos*	Number of Gait Cycles Before Data Cleaning**	Number of Gait Cycles After Data Cleaning**
NM	30	59	266	177
EL	15	30	192	154
MD	20	40	278	172
SV	15	30	297	222
Total	80	79	1033	725

The number of subjects, gait videos, and gait cycles for each osteoarthritis severity group: Normal (NM), Early (EL), Moderate (MD), and Severe (SV). Data include both walking directions and both legs. Gait cycles were filtered to remove incomplete or noisy segments.Videos include both Left-to-Right and Right-to-Left directions, *Gait cycles include right and left legs* *Including Left-to-Right and Right-to-Left directions. **Including right and left legs.

2.3. Classification

For the classification of gait patterns corresponding to varying severity levels of knee osteoarthritis (OA), we utilized an LSTM-FCN model [17]. This architecture combines the strengths of FCNs, known for their ability to extract robust spatial and temporal features from time-series data, with LSTM units, which effectively capture long-term dependencies in sequential data. The integration of LSTM layers into the FCN framework enhances the network's capacity to model intricate temporal dynamics present in gait cycles, thereby improving classification accuracy while maintaining a streamlined model structure.

The network was trained on pre-processed gait cycle data, which was carefully prepared to ensure compatibility with the LSTM-FCN architecture. The pre-processing pipeline focused on retaining critical temporal characteristics of the gait signals while mitigating noise, thereby facilitating effective learning of gait-related patterns associated with varying OA severity levels. The implementation of the LSTM-FCN

algorithm was carried out using the tsai framework [18], specifically designed for time-series analysis. The framework provided an efficient and streamlined environment for handling time-series data, enabling seamless integration of the LSTM-FCN architecture.

To ensure optimal performance, we conducted a systematic hyperparameter search using the Optuna library [19]. Optuna is an open source hyperparameter optimization framework designed to automate the search for optimal hyperparameters in machine learning models. It utilizes advanced algorithms, such as the Tree-structured Parzen Estimator (TPE), to efficiently explore high-dimensional search spaces and improve model performance.

The optimization process involved systematic tuning of a diverse set of hyperparameters, including the number of LSTM units, learning rate, dropout rate, and the number of convolutional filters. Some parameters were fixed based on practical considerations; for example, we set the number of epochs to 50 and the batch size to 64. The number of epochs was chosen to allow sufficient training time for model convergence without overfitting, while the batch size of 64 was selected based on common practice in deep learning tasks, providing a good balance between memory usage and training efficiency. The remaining hyperparameters were explored within the search space using Optuna. In the optimization process, we aimed to maximize the minimum F1 score for each fold and each class, ensuring that the model performed well across all classes and folds, not just on the average score. We used the TPE Sampler for the search and conducted 300 trials to find the optimal hyperparameter configuration.

2.4. Evaluation

To ensure a thorough assessment of our classification model's performance, we adopted a comprehensive evaluation strategy. Specifically, we designed two distinct data-splitting approaches, each aimed at evaluating different aspects of the model's generalization capabilities.

1. Random splitting: The dataset was randomly divided into training and test sets, allowing gait cycles from the same individual to appear in both sets, this means that gait data from the same individual (e.g., multiple gait cycles) could appear in both the training and test sets. This approach enables the model to leverage individual-specific patterns encountered during training. This evaluation setting primarily assesses the model's ability to generalize across different gait cycles of the same individual, rather than to generalize to entirely unseen subjects.
2. Subject-based splitting: All data from a given individual were exclusively assigned to either the training or test set, ensuring complete separation between subjects. This strategy provides a stringent evaluation of the model's ability to classify gait patterns from unseen individuals, which is critical for real-world applications.

For each data-splitting strategy, we implemented 5-fold cross-validation with stratified sampling to ensure robust and reliable performance evaluation. In the random splitting strategy, stratified sampling preserved the class distribution across all folds, minimizing the impact of class imbalance and ensuring a consistent representation of knee OA severity levels in both training and test sets. For the subject-based splitting strategy, stratified sampling was applied at the subject level, ensuring that each fold contained a similar proportion of individuals from each class while maintaining a strict separation of subjects between training and test sets. This setup enabled a fair evaluation of the model's ability to generalize to entirely unseen individuals while retaining the distribution of severity levels across folds. By combining 5-fold cross-validation with stratified sampling, we accounted for dataset variability and mitigated the risk of biased performance metrics due to uneven splits. This approach provided a comprehensive and consistent assessment of the model's performance under both data-splitting strategies.

Model performance was evaluated using multiple quantitative metrics, primarily classification F1-score and Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) score, to provide a rigorous assessment of the model's capability to distinguish between varying levels of knee OA severity. F1-score is the weighted average of precision and recall, providing a balance between the model's ability to correctly identify positive cases (precision) and its ability to capture all relevant positive cases (recall). It is especially useful when dealing with imbalanced datasets where one class may be underrepresented. ROC-AUC, on the other hand, measures the model's ability to discriminate between classes by evaluating the area under the receiver operating characteristic curve across different thresholds. It is computed using the One-vs-Rest (OVR) method for multiclass classification, where ROC-AUC is calculated for each class against all others, providing a comprehensive view of the model's performance across knee OA severity levels.

3. Results

In this study, the performance of the LSTM-FCN model was evaluated for classifying gait patterns associated with varying severity levels of knee OA. The model was trained using pre-processed gait cycle data derived from the 3D lower limb joint locations of both knee OA patients and healthy participants. Model performance was assessed through multiple metrics, including classification F1-score and ROC AUC score, ensuring a comprehensive evaluation of its ability to differentiate between knee OA severity levels.

Hyperparameter optimization was performed using the Optuna framework, with the objective of maximizing the minimum F1 score across classes and folds. The final model was evaluated using stratified 5-fold cross-validation under both random and subject-based data-splitting strategies. The search space and final selected hyperparameters resulting from the Optuna optimization loop are shown in Table 2. Notably, the subject-based split required a larger hidden neuron size and more LSTM layers, suggesting a need for higher model complexity to generalize across unseen individuals.

We evaluated the performance of our model using 5-fold cross-validation with the optimal hyperparameters identified through the Optuna optimization process. The results are presented in Table 3, which includes class-specific F1 scores, weighted F1 scores, and ROC AUC scores for two experimental setups. The reported results represent the mean and standard deviation across the 5 folds, providing a comprehensive view of the model's performance and its consistency. Detailed results for each individual fold are provided in Appendix A, allowing for a more granular examination of the model's performance across different subsets of the data. Appendix B shows the distribution of subject sample counts per class across folds and split strategies, while Appendix C presents the additional evaluation metrics calculated by

Table 2
Optuna search space and best-performing hyperparameters for random and subject-based splits.

Hyperparameter	Search Values (min, max)	Random Split	Subject-based Split
Learning rate	1e-5, 1e-2	8.51e-3	3.24e-3
Hidden neuron size	10, 100	10	63
LSTM layers	1, 3	2	3
Cell dropout	0, 1	0.69	0.71
LSTM dropout	0, 1	0.63	0.54
FC dropout	0, 1	0.12	0.05
Convolutional layers*	[8, 16, 8], [16, 32, 16], [32, 64, 32]	[32, 64, 32]	[32, 64, 32]

The search ranges and optimal hyperparameter values identified using Optuna for the LSTM-FCN model under two different data-splitting strategies. The optimization objective was to maximize the minimum F1 score across all classes and cross-validation folds. Convolutional layers were selected from predefined configurations. *Categorical values are selected from the given list.

Table 3
Classification performance by OA severity group and data-splitting strategy (mean ± standard deviation).

Split Strategy	Class F1 Scores				Weighted F1 Scores	ROC AUC
	NM	EL	MD	SV		
Random	0.94 ± 0.03	0.84 ± 0.06	0.85 ± 0.03	0.97 ± 0.01	0.91 ± 0.02	0.98 ± 0.01
Subject Based	0.87 ± 0.07	0.62 ± 0.09	0.63 ± 0.12	0.87 ± 0.07	0.76 ± 0.06	0.93 ± 0.03

The class-wise F1 scores, weighted F1 scores, and ROC AUC scores of the LSTM-FCN model for both random and subject-based data-splitting strategies. Results are averaged over 5-fold cross-validation. NM = Normal; EL = Early OA; MD = Moderate OA; SV = Severe OA.

aggregating the outputs from each fold.

The model performed well with random splitting (F1: 0.91, AUC: 0.98), particularly for Severe OA, but showed reduced performance with subject-based splitting (F1: 0.76, AUC: 0.90).

Fig. 4 presents the confusion matrices for the classification performance of (a) Random Splitting and (b) Subject-based Splitting models of severity levels. In both matrices, most values were concentrated along the diagonal, indicating that both splitting models were successful in correctly classifying the severity levels. Random splitting achieved fewer misclassifications across severity levels, suggesting better classification performance. Subject-based splitting showed a noticeable number of misclassifications, particularly with EL being confused with MD (73 instances), and SV being misclassified as MD (31 instances). Subject-based splitting had a higher confusion rate, particularly between EL and MD levels, but classified NM and SV, with minimal errors.

4. Discussion

This study demonstrates the efficacy of combining markerless motion capture with deep learning to classify knee OA severity based on gait kinematics. Using an LSTM-FCN model, we achieved strong classification performance, highlighting the potential of this non-invasive and accessible approach for evaluating functional impairments linked to OA progression. Our findings reveal distinct gait patterns corresponding to different Kellgren–Lawrence severity grades, with the model effectively capturing both temporal dynamics and spatial variations in gait cycles. These results underscore the value of gait biomechanics as a supportive tool to radiographic imaging in assessing disease severity.

Random splitting strategy enabled the model to learn from a diverse set of individuals across training and test folds, resulting in higher accuracy and generalization within the dataset. The random-split results are included to approximate an intra-subject evaluation, where samples from the same individuals appear in both the training and test sets. Given the presence of intra-subject correlation and the associated risk of data leakage, this evaluation does not provide a valid estimate of the model's generalizability and should not be interpreted within a clinical context. While the random-splitting strategy resulted in higher performance metrics, these findings are reported solely for exploratory purposes. On the other hand, subject-based splitting strategy assessed the model's ability to generalize to completely unseen individuals, offering a more realistic measure of real-world performance. The lower accuracy suggested that the model may have overfitted individual gait characteristics rather than learning universal OA severity patterns. Therefore, the subject-based cross-validation offers a more reliable and clinically meaningful assessment, as it evaluates the model's ability to generalize to entirely new individuals. While the random split can reveal how well the model captures within-subject consistency and repeated gait patterns, it may also introduce bias by allowing subject-specific features to influence both sets, potentially leading to overestimated performance. Therefore, we emphasized that the subject-based split provides a more rigorous assessment of the model's generalization to unseen individuals and serves as our primary evaluation metric. The random split results were included as a complementary analysis to explore model behavior in the presence of subject-level continuity.

Across both strategies, Severe OA (SV) and Normal (NM) were consistently well classified, indicating distinct gait characteristics, whereas Early OA (EL) and Moderate OA (MD) had higher misclassification rates—particularly in subject-based splitting—suggesting substantial feature overlap that complicates classification. The superior performance in random splitting highlighted a risk of overfitting when the same subjects appear in both training and test sets, while subject-based splitting emphasized the challenge of inter-subject gait variability.

This study leveraged markerless pose estimation, ensuring precise and consistent joint location sequences, which were critical in classifying OA severity. By eliminating the need for reflective markers or controlled lab environments, our approach enhances the accessibility and scalability of gait analysis. Prior research primarily relied on radiographic assessments or marker-based gait analysis. For example, in

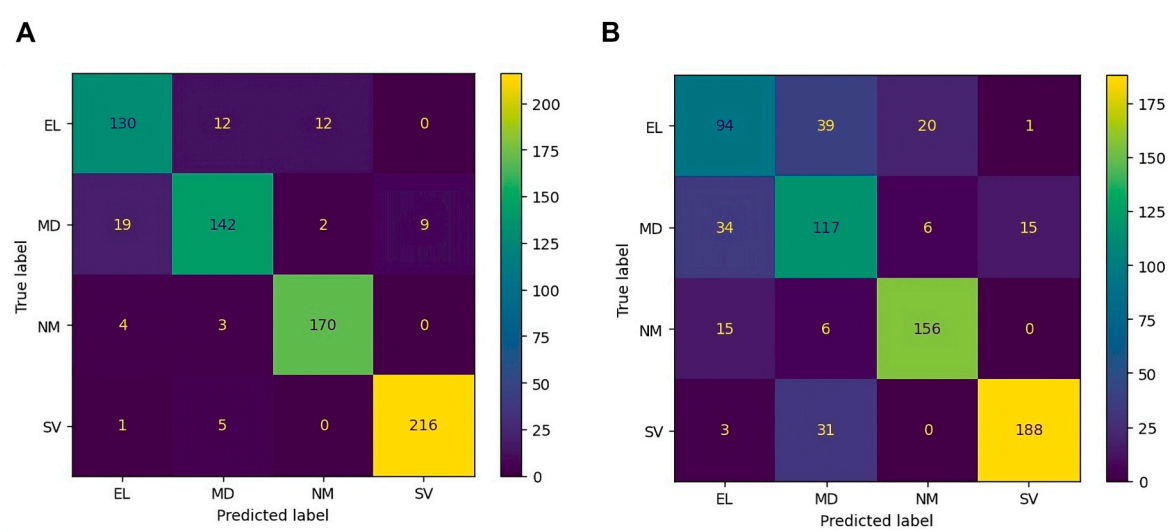

Fig. 4. Confusion matrices for LSTM-FCN knee osteoarthritis severity classification: (a) Random split; (b) Subject based split.

Ref. [12] Kwon et al. identified gait features correlated with OA severity using machine learning and marker-based systems. While their findings validated gait metrics as predictive indicators, their methods required specialized equipment, limiting real-world applicability. In contrast, our markerless approach democratizes biomechanical assessments, aligning with recent computer vision and AI-driven innovations. Moreover, compared to earlier approaches such as Elbaz et al., who used spatio-temporal gait parameters from pressure-sensitive walkways to classify knee OA with moderate accuracy [20], our model incorporates lower-body kinematic trajectories obtained via markerless motion capture and processed using a deep learning architecture. This allows for more nuanced pattern recognition across the entire gait cycle. While Elbaz et al.'s method is grounded in clinically interpretable variables such as stride length and cadence, it requires controlled laboratory settings, dedicated hardware and manual preprocessing steps. Our method, by contrast, provides a scalable alternative that maintains high classification performance and captures joint-level biomechanical deviations relevant to KOA severity.

The markerless systems, when combined with advanced deep learning frameworks, can capture clinically relevant features traditionally measured using high-end motion capture setups. Importantly, this extends the reach of gait analysis beyond research laboratories and into more routine clinical settings. Our approach, therefore, holds promise not only for automated OA severity classification, but also for early screening, functional monitoring, and personalized rehabilitation planning—particularly in resource-limited environments [21–23]. Unlike static radiographic assessments, gait-based evaluations capture functional impairments in real time, offering a dynamic perspective of disease progression.

Although the findings are encouraging, this study has the following limitations. First, sole reliance on gait kinematics—excluding potential biomarkers such as muscle strength, joint loading, and biochemical markers. Second, the lack of multimodal integration, which could enhance classification accuracy and provide a comprehensive OA assessment and the lack of longitudinal data, limiting insights into the model's ability to predict disease progression over time. Third, it is important to note that the dataset used in this study comprises an Indian population with a lower average height (~1.56 m) compared to Western populations such as the MOST cohort (~1.69 m), which may influence gait characteristics and limit the generalizability of our findings across different demographic groups. Additionally, the OA severity was graded for the most affected knee, but the dataset did not specify whether the condition was unilateral or bilateral. As we included gait cycles from both limbs as input for the model, this approach does not account for potential differences in gait patterns between unilateral and bilateral OA, which could impact the overall results. The exploration of alternative pose estimation algorithms and a more detailed examination of the preprocessing pipeline could be pursued as separate areas of future research. Additionally, external validation is crucial for confirming the generalizability of our model, but the lack of a publicly available kinematics or video dataset with knee OA severity levels prevented the creation of a fully independent validation set. As a solution, we utilized k-fold cross-validation to improve the robustness and reliability of our results; however, we plan to develop a larger, clinically verified gait dataset to enable comprehensive external benchmarking and further support the clinical translation of our model. While the window size of 5 was selected based on empirical observations, future work should explore quantitative validation techniques to further optimize this parameter and assess its impact on downstream model performance. Although the LSTM-FCN model achieved strong performance in classifying OA severity based on spatiotemporal gait trajectories, this study did not include a dedicated analysis of feature importance. In the context of deep learning, especially for clinical applications, interpretability remains a key consideration. Standard feature ablation techniques are challenging to implement given the sequential and high-dimensional structure of the input data, which consists of time-series joint trajectories rather than independent scalar features. Nonetheless, methods such as joint-wise perturbation, offer promising avenues for identifying the most influential joints and gait phases. Future work incorporating these techniques will be vital for improving model interpretability, identifying potential sources of noise, and fostering greater clinical confidence in AI-driven gait analysis. Moreover, although this study demonstrates the feasibility of using markerless motion capture for OA severity classification, the absence of clinical validation for certain components of the processing pipeline must be acknowledged. Specifically, while the AlphaPose and HybrIK framework has shown strong performance on general 3D pose estimation benchmarks, its accuracy in capturing clinically relevant frontal and transverse plane movements—such as knee varus/valgus or axial rotation—remains uncertain. In this study, the joint angles were derived in the sagittal plane and used solely for outlier filtering during preprocessing. However, we recognize that errors in joint estimation, even at the preprocessing stage, can potentially affect input quality and model outcomes. Additionally, although we adapted a previously validated gait event detection algorithm, we did not conduct a formal accuracy evaluation of this method within our markerless framework due to the lack of ground truth force plate data. Future studies should include rigorous validation of both 3D pose estimation accuracy and gait event detection performance. Moreover, future research should incorporate multimodal data, integrating clinical and functional biomarkers to improve model accuracy and deepen insights into OA mechanisms. In particular, combining markerless gait kinematics with demographic and clinical variables such as age, gender, and BMI may enhance both predictive performance and clinical applicability. Longitudinal studies will also be crucial for assessing the model's predictive power in tracking OA progression and treatment outcomes. Finally, OA severity labels were assigned based on the most affected knee, whereas gait data were collected bilaterally. Since gait reflects the integrated function of both lower limbs, this labeling approach may not fully capture the biomechanical variability between unilateral and bilateral OA presentations. As a result, the current classification framework may limit the model's interpretability and its applicability to diverse clinical scenarios. Future studies should explore classification schemes that account for bilateral KL grade combinations (e.g., EL-MD, MD-SV) to more accurately reflect the spectrum of gait adaptations in individuals with knee OA.

5. Conclusions

This study highlights the potential of markerless motion capture and deep learning for knee osteoarthritis (OA) severity classification based on gait kinematics. Using an LSTM-FCN model, we effectively classified OA severity levels, underscoring the potential of gait biomechanics as a functional modality that enhances image-based assessments provided by radiographic imaging. While both random and subject-based data-splitting strategies were explored, we emphasize that subject-based cross-validation provides the most reliable estimate of the model's ability to generalize to new individuals and therefore forms the basis for clinical interpretation. Performance metrics obtained through random splitting, where gait cycles from the same subjects may appear in both training and test sets, are included for exploratory purposes only. These results may be inflated due to subject-specific biases and should not be interpreted as evidence of generalizability. Additionally, the random-split approach does not simulate longitudinal disease tracking, as KL grades in this dataset are static and do not reflect temporal progression. Data-splitting strategies significantly influenced performance: random splitting achieved higher accuracy but risked overfitting, while subject-based splitting provided a more realistic assessment, revealing challenges in generalizing to unseen individuals. Severe OA (SV) and Normal (NM) were well classified, whereas Early (EL) and Moderate OA (MD) showed greater misclassification due to overlapping gait patterns. While markerless gait analysis proves to be a non-invasive, scalable assessment tool, future work should focus on enhancing feature extraction,

integrating multimodal biomarkers, and exploring domain adaptation to improve generalizability. Longitudinal studies will be key in assessing disease progression and treatment outcomes.

CRediT authorship contribution statement

Engin Kaya: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hülya Şirzai:** Writing – review & editing, Conceptualization. **Güneş Yavuzer:** Writing – review & editing, Supervision, Conceptualization. **Hande Argunsah:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization.

Informed consent statement

Not applicable.

Institutional review board statement

Not applicable.

Data availability statement

The gait dataset used in this study is publicly available at Mendeley Data: https://data.mendeley.com/datasets/44pfnysy89/1. It can be accessed freely for research purposes and includes detailed documentation. The processed datasets generated during preprocessing steps, as well as the specific data splits used for training and evaluation, are available from the corresponding author upon reasonable request.

Ethics statement

This study utilized gait data sourced from the publicly available dataset referenced in text, entitled "A vision-based gait dataset for knee osteoarthritis and Parkinson's disease analysis with severity levels". Therefore this research does not need an Ethics approval.

Declaration of competing interest

The authors declare no conflicts of interest.

Abbreviations

OA	Osteoarthritis
NM	Normal
EL	Early
MD	Moderate
SV	Severe
LSTM	Long Short-Term Memory
FCN	Fully Convolutional Network

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compbiomed.2025.110647.

Appendix A

Table 1
Class and overall results for each fold.

	Fold	Class F1 Scores				Weighted F1 Scores	ROC AUC
		NM	EL	MD	SV		
Random	1	0.931507	0.833333	0.882353	0.966292	0.909548	0.986346
	2	0.947368	0.813559	0.835821	0.977273	0.901833	0.979788
	3	0.929577	0.818182	0.825397	0.977778	0.895241	0.985482
	4	0.985507	0.950820	0.882353	0.956522	0.944908	0.996293
	5	0.916667	0.806452	0.823529	0.954545	0.883020	0.980752
Subject Based	1	0.843373	0.760563	0.821918	0.950495	0.855704	0.970575
	2	0.767123	0.549020	0.611111	0.815789	0.698973	0.898677
	3	0.894737	0.575758	0.681319	0.792453	0.740017	0.934721
	4	0.911765	0.576923	0.526316	0.901099	0.753054	0.925308
	5	0.949153	0.633333	0.527778	0.895238	0.773633	0.933942

Appendix B

Table 1
Distribution of subject sample counts per class across folds and split strategies.

Split Type	Fold	EL	MD	NM	SV
Random Split	**Fold 1**	31	34	36	44
	Fold 2	30	35	36	44
	Fold 3	31	35	35	44
	Fold 4	31	34	35	45
	Fold 5	31	34	35	45
Subject-Based Split	**Fold 1**	33	38	41	52
	Fold 2	25	36	38	37
	Fold 3	31	42	38	32
	Fold 4	31	27	31	45
	Fold 5	34	29	29	56

Appendix C

Table 1
Additional evaluation metrics for random and subject-based splits were calculated by aggregating the outputs from each fold.

Metric	Random Split	Subject-Based Split
Accuracy	0.9076	0.7655
PPV	0.9068	0.7720
NPV	0.9716	0.9240

PPV: Positive Predictive Value, NPV: Negative Predictive Value for the LSTM-FCN model under both random and subject-based data-splitting strategies.

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